

TrES-5b

R-1981

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_130813 · created 2026-08-07T00:47:18+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456517.8797 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.179 +/- 0.051
Transit depth (Rp/Rs)^2	3.21 +/- 1.82 [%]
Orbital Inclination (inc)	81.66 +/- 2.79
Airmass coefficient 1 (a1)	0.898 +/- 0.047
Airmass coefficient 2 (a2)	0.092 +/- 0.038
Scatter in the residuals of the lightcurve fit is	5.4 %
Best Comparison Star	#1 - [123, 130]
Optimal Aperture	2.77
Optimal Annulus	10.08
Transit Duration (day)	0.051 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.179 ± 0.051 vs 0.142 (deviation 0.66σ)
Duration fit vs published	0.051 d vs 0.0758 d (deviation 0.78σ)
Mid-time O–C	-3.5 min
Depth SNR	3.2
Residual scatter	5.4 %

Light curve

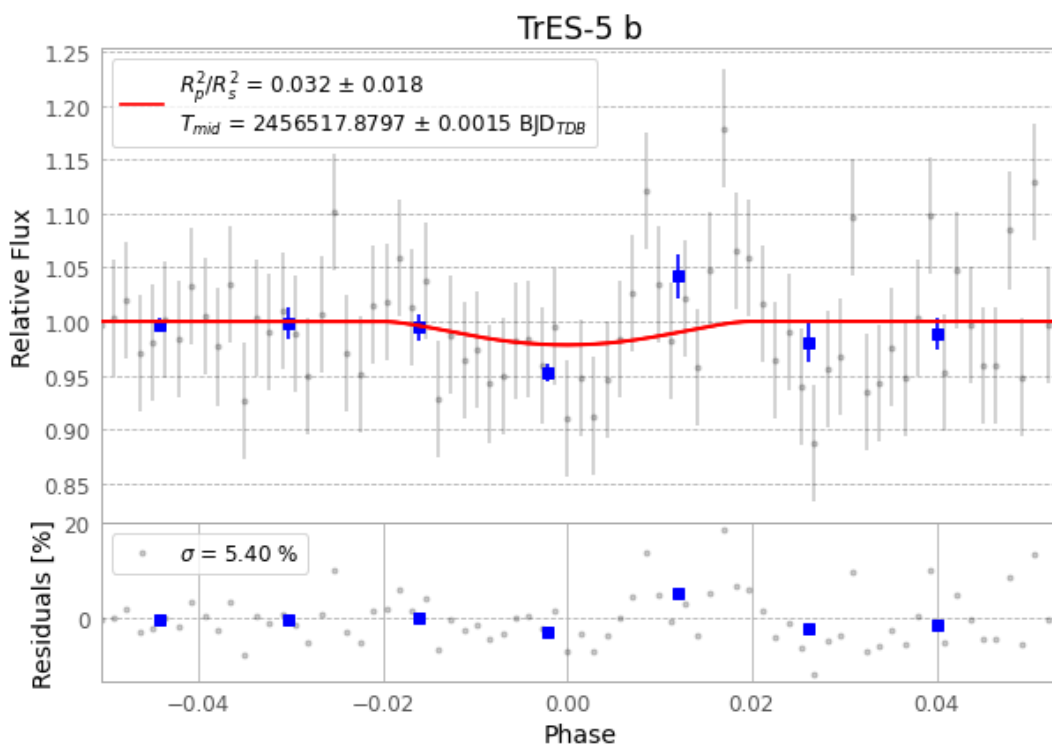


Figure 1 — FinalLightCurve_TrES-5 b_2013-08-13.png (SHA-256 55b8b3dea0b51e3e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [417, 273], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3853fbf14af9ce53...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

77 FITS frames (51 MB), TRES-5130813071512.FITS → TRES-5130813110312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-130813.log (6f4c08b85311...), FinalLightCurve_TrES-5 b_2013-08-13.png (55b8b3dea0b5...), FinalLightCurve_TrES-5 b_2013-08-13.csv (60a3f2be65d8...)

Compute resources

Wall time 188.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 00:47Z.